The Electric Potential as Rope Tension

The Coulomb Potential as a Mechanical Tension Field — a Classical In-Domain Result

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Abstract

This paper reports the rope framework's mechanical reading of the electric potential: the scalar potential A_0 is identified with a tension field of the rope network, so that the Coulomb 1/r potential of a point charge arises as the static tension configuration sourced by a linked structure. This is a classical, in-domain result — the potential is given a literal mechanical meaning consistent with the electromagnetic constants reported elsewhere in the corpus. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants.

Scope of this paper

CLAIM: the electric scalar potential is a mechanical tension field of the rope network; the Coulomb 1/r potential is the static tension sourced by a point charge (a linked structure).

NOT CLAIMED: anything about quantum electrodynamics or the quantum sector. A classical electrostatics result, in-domain. Status: Modeled.

1. Potential as Tension

In the rope picture, the electromagnetic potential is not an abstract field but the tension state of the rope network. The scalar potential A_0 corresponds to the static tension, and a point charge — a linked rope structure — sources a tension field that falls off as 1/r, reproducing the Coulomb potential.

$$A_0(r) \propto 1/r \quad (1.1)$$ — Coulomb, as sourced tension

2. Consistency With the EM Sector

This mechanical reading is consistent with the electromagnetic constants computed in the corpus (ε_0, Z_0, α) and with the identification of charge as linking number: a charge is a link, and its potential is the tension that link sources. The account is classical throughout and sits squarely in the programme's in-domain region.

Status

Electric potential = mechanical tension field — Modeled (in-domain).

Coulomb 1/r from sourced tension — Modeled.

Quantum electrodynamics — out of scope (not addressed).

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.